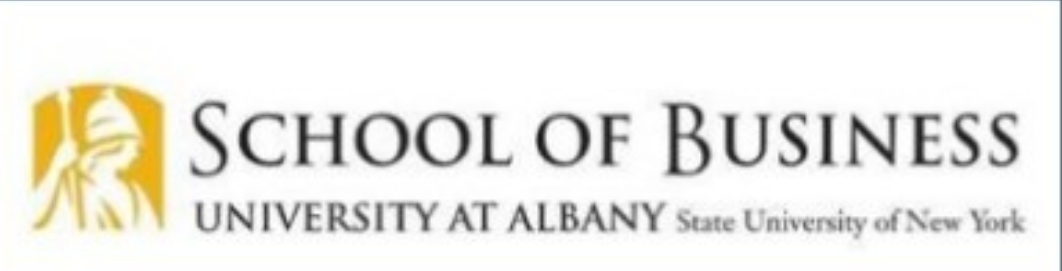


DSM-5-Guided Multimodal Behavioral Biomarkers for Depression & PTSD Assessment

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1 The Problem

AI can flag depression and PTSD, but rarely shows why.

280M+ with depression Binary output Post-hoc XAI

Common, disabling, and often co-occurring

Depression affects more than 280 million people, and PTSD frequently co- occurs with it. Clinical interviews remain essential, but they are time-intensive, episodic, and subject to recall bias and inter-rater variability.

Predictions without clinical structure

Most systems output a single participant-level label. SHAP and LIME show which features matter, but not which DSM-5 symptom they reflect, or whether signs point to depression, PTSD, or both.

2 Our Approach

A DSM-5-guided multimodal framework for depression & PTSD

We integrate audio and visual behavioral biomarkers with structured psychiatric knowledge, organizing model evidence by clinically meaningful symptom domains instead of a single undifferentiated outcome. Depression is modeled first; PTSD extends the same pipeline.

THREE PROGRESSIVE REPRESENTATIONS

Behavioral baseline	Temporal dynamics	DSM-5 criteria
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✓ Participant-level prediction preserved

Intended to support, not replace, clinical interviews and professional diagnosis.

3 Datasets & Features (Depression)

01 D-Vlog · naturalistic
701 quality-filtered YouTube vlogs (370 depression, 331 non-depression), self- recorded and unscripted.

03 Audio · 16 kHz
OpenSMILE, COVAREP, Librosa: pitch, loudness, MFCCs, spectral, glottal & voice-quality, word counts.

02 E-DAIC · clinical
275 semi-structured interviews, PHQ-8 labels; 66 depression-labeled sessions added (37 / 12 / 17 split).

04 Visual · OpenFace 2.0
Action Units, gaze, blinks, 3D head pose, smile intensity; frames with confidence < 0.9 removed.

4 Three-Phase Framework (Depression)

(1) PHASE	(2) REPRESENTATION	(3) SELECTION & MODEL
Phase I Behavioral baseline	Toolkit-derived descriptors No temporal transforms	→ XGBoost + SHAP → 38 audio · 43 visual
Phase II Temporal dynamics	V1: gaze avoidance, prosody V2: spectral tilt, AU entropy	→ VIF + CatBoost: 41 · 35 → Fisher ratio: 12 · 35
Phase III DSM-5 criterion-level	6 DSM-5 symptom domains Forward selection on F1	→ Per-domain CatBoost → 2nd-stage fusion model

Selected descriptor counts shown as audio · visual. Session features use mean, std, min, max (+ median).

5 DSM-5 Mapping (Depression)

DSM-5 SYMPTOM DOMAIN	AUDIO EVIDENCE	VISUAL EVIDENCE
A1 Depressed mood	Prosody, spectral & harmonic	Facial activity, smile dynamics
A2 Anhedonia	Vocal variability, loudness	Expressivity, smiling
A5 Psychomotor changes	Speech pacing, word count	Head pose, gaze, AU timing
A6 Fatigue	Vocal intensity, speech rate	Facial activity, blinks
A7-8 Worthlessness & concentration	Verbal production statistics	Gaze, head movement

Criteria requiring self-report (e.g., sleep, appetite, suicidality) are not modeled independently.

6 Framework Overview (Depression)

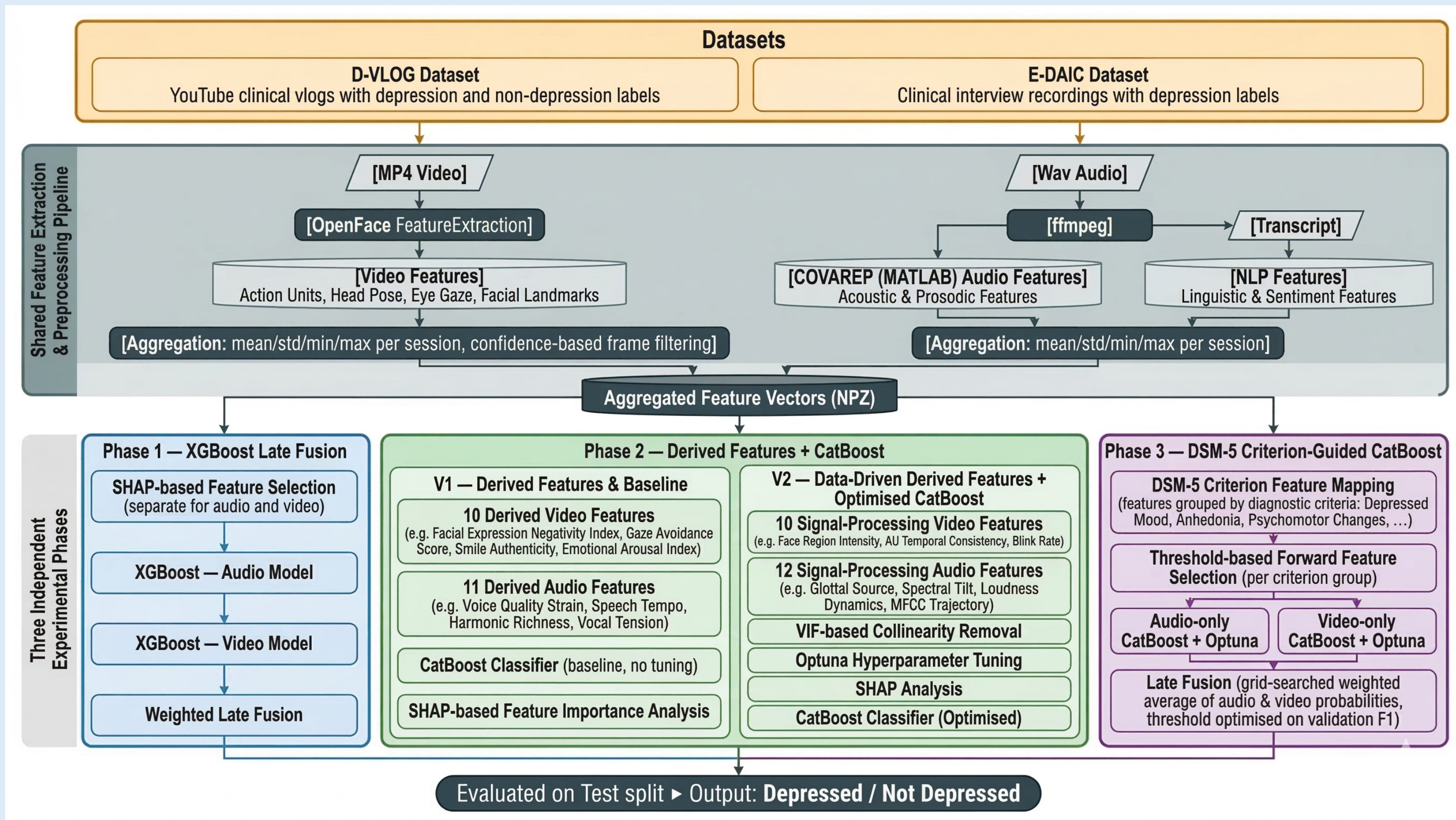


Fig. 1: Depression pipeline framework

7 Experimental Results (Depression)

0.765 Phase III fusion F1 (DSM-5)	0.833 Phase III fusion recall	0.804 Best baseline: Phase I audio
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Depression-class test F1 by phase

Phase	Audio	Video	Fusion
I · Baseline	.804	.716	.796
II · Temporal V1	.704	.687	.739
II · Temporal V2	.697	.709	.709
III · DSM-5	.733	.671	.765

Fusion F1 by phase

Phase I	0.796
Phase II-V1	0.739
Phase II-V2	0.709
Phase III	0.765

Audio leads: speech carries the strongest signal; video adds complementary evidence.

Temporal features shift modality balance but do not uniformly raise F1.

DSM-5 fusion beats both single modalities in Phase III while adding clinical structure.

8 Prior Work Comparison (Depression)

MODEL · DATASET	F1	EXPL.	DSM-5
Yoon et al. · D-Vlog	0.63	No	No
Zheng et al. · D-Vlog	0.65	No	No
Cheong et al. · D-Vlog	0.70	No	No
Williamson et al. · DAIC	0.72	No	No
Alhanai et al. · DAIC	0.77	No	No

This work · D-Vlog + E-DAIC	0.765	Yes	Yes
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LIMITATIONS

- Audio-visual signals cannot capture internal, unobservable symptoms
- Naturalistic recording conditions affect feature reliability
- DSM-5 mapping needs symptom-level annotated validation

CONCLUSION & FUTURE WORK

DSM-5 structure organizes evidence without eliminating predictive capability. Next: linguistic and longitudinal modeling, external validation, and clinician-centered evaluation.

9 In Progress: PTSD Track

We are extending the same audio-visual pipeline to PTSD using the **PTSD in the Wild** dataset (video and audio recordings). An earlier binary PTSD model (XGBoost, early fusion, K-fold validation) showed combined audio + video outperforming single modalities.